

Why Banik and Zhao Were Right

External-field structure and a central tidal
consistency relation in filtered MOND

Carl Zimmerman

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Abstract

A modified-gravity law cannot be tested from an isolated spherical force curve alone: its response to an ambient field also matters. We revisit the external-field linearization of QUMOND and study an explicitly varied, spatially filtered specialization of its two-potential action. The Newtonian source is left unsmoothed; variation places an adjoint filter outside the nonlinear phantom-density source. For a spherical source perturbing a constant, nonzero constitutive background, isotropic filtering produces a finite central phantom-potential Hessian whenever the stated spectral integral converges. Its trace D and axial quadrupole Q_2 obey a division-free consistency identity independent of source mass and filter amplitude. For the inverse partner of $\mu(y) = 1 - e^{-y}$, the ratio is $Q_2/D = 2y/\{5[3 + 3(y-1)e^{-y} - y]\}$ where $D \neq 0$. Symbolic identities and independent quadrature over 324 cases check this result, with a deliberately anisotropic control violating it. The contribution is a conditional central-response corollary, not a new MOND architecture, an empirical detection, or a completed relativistic theory.

1 What the title means

Banik and Zhao derived the anisotropic external-field-dominated response of AQUAL and QUMOND [1]. In QUMOND their equation (24) contains both the value of the interpolation function and its logarithmic derivative. Suppressing or retaining a scalar force enhancement therefore does not, by itself, specify the directional response. This conditional mathematical point is the sense in which the title says they were right; we do not adjudicate their observational claims or any wide-binary controversy.

The parent two-potential action and external-field asymptotics are due to Milgrom [2]. His later generalization explicitly allows a possibly nonlocal functional of the Newtonian potential and investigates extra-scale screening [3]. Thus neither an additional length scale nor a nonlocal QUMOND action is claimed here as a new architecture.

Our narrower question can fail: once an isotropic filter changes the amplitude of the central response, can it tune the isotropic and anisotropic parts independently? In the linear regime below, the answer is no. The resulting relation provides a joint consistency test of this candidate, provided both components can eventually be measured.

2 One static action and its variations

Let u be an auxiliary Newtonian potential and Φ the potential felt by nonrelativistic matter. Fix $a_0 > 0$ and a real, normalized, isotropic convolution S on space. Its Fourier multiplier is $\sigma(k)$, with $\sigma(0) = 1$. The filter is self-adjoint under the boundary prescription and commutes with spatial derivatives. A Gaussian example is $S = \exp(\xi^2 \Delta/2)$, $\sigma(k) = \exp(-\xi^2 k^2/2)$.

Use the exact inverse constitutive pair

$$\mu(y) = 1 - e^{-y}, \quad s = y\mu(y), \quad \nu(s) = y/s, \quad y > 0. \quad (1)$$

The inverse is unique since $ds/dy = 1 + (y-1)e^{-y} > 0$ for $y > 0$. The exponential function itself is not new; it appears explicitly in earlier Solar-System MOND work [4]. Define

$$\mathcal{G}(y) = y^2 + 2(1+y)e^{-y} - 2, \quad \mathcal{G}'(y) = 2s, \quad (2)$$

$$q(s^2) = 2sy - \mathcal{G}(y) - s^2, \quad q'(s^2) = \nu(s) - 1. \quad (3)$$

Here q' differentiates its single argument, not s . Consider

$$I = \int dt d^3x \left\{ \mathcal{L}_{m,\text{kin}} - \rho_b \Phi - \frac{2\nabla\Phi \cdot \nabla u - |\nabla u|^2 - a_0^2 q(|\nabla Su|^2/a_0^2)}{8\pi G} \right\}. \quad (4)$$

This is a spatially nonlocal, nonrelativistic action, not an Einstein-Hilbert completion. It specializes Milgrom's functional construction with $P[u] = |\nabla u|^2/a_0^2 + q(|\nabla Su|^2/a_0^2)$ [3].

Set $f(p) = [\nu(|p|/a_0) - 1]p$. Compatible fixed-boundary variations give

$$\frac{\delta I}{\delta \Phi} = \frac{\Delta u}{4\pi G} - \rho_b = 0, \quad (5)$$

$$\frac{\delta I}{\delta u} = \frac{\Delta \Phi - \Delta u - S^* \nabla \cdot f(\nabla Su)}{4\pi G} = 0. \quad (6)$$

Consequently,

$$\boxed{\Delta u = 4\pi G \rho_b, \quad \Delta \Phi = 4\pi G \rho_b + S^* \nabla \cdot f(\nabla Su).} \quad (7)$$

In particular, $\int f \cdot \nabla S \delta u = - \int \delta u S^* \nabla \cdot f$: the outer filter is required by variation. Matter-coordinate variation gives acceleration $-\nabla\Phi$; it does not establish a four-dimensional matter Ward identity. The unsmoothed Newtonian term is retained exactly.

For infinite space these are variational field equations with fixed asymptotics or background subtraction, not a claim of finite unsubtracted action. At $S = 1$ they reduce to QUMOND. Finite filtering preserves the chosen constitutive function, *not* the original universal AQUAL equation for Φ . Even unfiltered QUMOND and AQUAL are not generally identical outside the appropriate symmetric configurations.

3 A central-tide consistency identity

Take a uniform background $p_e = \nabla S u_e = a_0 s_e \hat{e}$, with $s_e > 0$. The filtered source perturbation must be small relative to $|p_e|$ throughout the response region. Define

$$A = \nu(s_e) - 1, \quad B = s_e \nu'(s_e), \quad Df(p_e)_{ij} = A\delta_{ij} + B e_i e_j. \quad (8)$$

All following source responses are first order in source amplitude, at fixed background. For a point-source perturbation of mass M , (7) gives

$$\hat{\Phi}_{\text{ph}}(\mathbf{k}) = -\frac{4\pi GM}{k^2} |\sigma(k)|^2 [A + B(\hat{\mathbf{k}} \cdot \hat{\mathbf{e}})^2], \quad k \neq 0, \quad (9)$$

where Φ_{ph} subtracts the Newtonian source potential and the prescribed external background. The inverse Fourier measure is $d^3k/(2\pi)^3$. Choose a decaying perturbation at infinity and no additional external tidal harmonic. Neither the potential constant nor the background is determined by inverting the $k \neq 0$ equation.

Proposition 1. *Assume isotropic S , a spherical source, the nonzero uniform background and linear regime above, and a convergent central spectral integral. For a point source put $I_\sigma = \int |\sigma(k)|^2 d^3k/(2\pi)^3$. The central phantom-potential Hessian is*

$$H_{ij} = \partial_i \partial_j \Phi_{\text{ph}}(0) = \frac{4\pi GM I_\sigma}{15} [(5A + B)\delta_{ij} + 2B e_i e_j]. \quad (10)$$

With $H_{\parallel} = e_i H_{ij} e_j$, $H_{\perp} = (\text{tr } H - H_{\parallel})/2$, $Q_2 = H_{\perp} - H_{\parallel}$ and $D = \text{tr } H$, one has

$$\boxed{(15A + 5B)Q_2 + 2BD = 0.} \quad (11)$$

For a finite spherical source, replace MI_σ everywhere by $J_\rho = \int \hat{\rho}_b(k) |\sigma(k)|^2 d^3k/(2\pi)^3$, assuming absolute convergence. The same identity holds, even if $J_\rho = 0$.

Proof. Two derivatives in (9) contribute $-k_i k_j$, canceling the potential's minus sign. Isotropy separates the radial integral from

$$\langle n_i n_j \rangle = \frac{\delta_{ij}}{3}, \quad \langle n_i n_j (n \cdot \hat{\mathbf{e}})^2 \rangle = \frac{\delta_{ij} + 2e_i e_j}{15}. \quad (12)$$

These angular moments give (10). Its axial difference and trace are

$$Q_2 = -\frac{8\pi GM I_\sigma B}{15}, \quad D = 4\pi GM I_\sigma \left(A + \frac{B}{3} \right). \quad (13)$$

Substitution proves (11). A spherical finite source changes only the radial integral, establishing the final statement. $\square$

4 The exponential prediction and its interpretation

Write $s_e = y(1 - e^{-y})$ and $\lambda(y) = 1 + (y - 1)e^{-y}$. Differentiating the inverse relation, rather than assigning its response, yields

$$A = \frac{e^{-y}}{1 - e^{-y}}, \quad B = -\frac{ye^{-y}}{(1 - e^{-y})\lambda(y)}. \quad (14)$$

The result for the exponential constitutive law is therefore

$$\boxed{\frac{Q_2}{D} = \frac{2y}{5[3\lambda(y) - y]}} \quad (D \neq 0). \quad (15)$$

Mass, isotropic filter shape and smoothing length cancel. At $y = 1$ the ratio is $1/5$; as $y \rightarrow 0^+$ it approaches $2/25$. The latter is generic for a regular deep-MOND inverse asymptotic $\nu(s) \sim s^{-1/2}$ with its corresponding derivative asymptotic, not a unique exponential fingerprint. Taking this limit while retaining external dominance also requires reducing the source amplitude; the exactly zero-background problem is different.

For reference, the point-source amplitudes are

$$I_{\text{Gaussian}} = \frac{1}{8\pi^{3/2}\xi^3}, \quad I_{\text{Helmholtz}} = \frac{1}{8\pi\xi^3}, \quad (16)$$

for $\sigma = e^{-\xi^2 k^2/2}$ and $\sigma = (1 + \xi^2 k^2)^{-1}$ respectively. A general spherical density need not give positive J_ρ for every real isotropic filter. Nonnegative real-space Gaussian or Helmholtz kernels and nonnegative density secure positivity for a nonzero source. The tested finite sources are Gaussian. Positivity is not needed for the identity.

Point-source feature	Root equation	Computed y
$H_{\parallel} = 0$	$3y = 5\lambda(y)$	1.8907779792
$D = 0$	$y = 3\lambda(y)$	3.2602387018
$H_{\perp} = 0$	$y = 5\lambda(y)$	5.1228414320

At the middle root the Hessian is finite; only the quotient in (15) is undefined. Equation (11) is the primary statement. Between the first and third roots the additional acceleration stretches parallel to the background and compresses perpendicular to it. This is not a claim that total gravity is repulsive: the physical tidal acceleration tensor is $-H$, and the ordinary Newtonian force remains.

Locally the anomalous potential has the expansion

$$\Phi_{\text{ph}}(x) = \Phi_{\text{ph}}(0) + \frac{Dr^2}{6} - \frac{Q_2}{2} \left[(\hat{e} \cdot x)^2 - \frac{r^2}{3} \right] + o(r^2). \quad (17)$$

Thus $D = 4\pi G\rho_{\text{ph}}(0)$ is isotropic central curvature, not total phantom mass. Our Q_2 convention matches the traceless coefficient in Hees et al., equation (6) [4]. The extra D term and the weak-source assumption make this a different prediction from a nonlinear solar quadrupole calculation; an existing pure-quadrupole fit cannot simply be read as a fit of both coefficients here.

5 Reproducible checks, not observational evidence

The companion calculation computes the directional Hessian by numerical angular integration rather than setting Q_2/D to its expected value. SymPy separately checks exact angular moments, the inverse derivative, the division-free identity and the deep limit. Finite differences of the nonlinear flux check the constitutive Jacobian independently.

Input	Tested range
Background y	0.03, 0.3, 1, 2.5, 4, 6
Filters	Gaussian, Helmholtz, $1/[1 + (\xi k)^4]$
Widths ξ	0.4, 1, 2.5 (common numerical units)
Gaussian source radii R	0, 0.3, 2 ($R = 0$: point source)
Directions	(0, 0, 1) and $(1, 2, 3)/\sqrt{14}$
Full combinations	$6 \times 3 \times 3 \times 3 \times 2 = 324$

The code sets $G = M = 1$ to calculate linear-response coefficients; this is an amplitude normalization, not a claim that each dimensional realization is external-field dominated. Finite sources use $\hat{\rho}_b(k)/M = \exp(-R^2 k^2/2)$.

The maximum normalized identity residual is 1.10×10^{-14} , and the maximum relative tensor discrepancy is 1.99×10^{-14} . The former divides the absolute left-hand side of (11) by $(|15A + 5B| + 2|B|)\|H\|_F$; the latter is $\|H_{\text{num}} - H_{\text{analytic}}\|_F / \|H_{\text{analytic}}\|_F$. These are floating-point agreement measures, not empirical error bars or certified continuum error bounds.

Gaussian and Helmholtz radial integrals are checked against exact normalizations. Increasing the polar quadrature from 24 to 40 nodes and rotating the field axis checks the angular implementation. An intentionally anisotropic spectral weight $1 + 0.8(\hat{\mathbf{k}} \cdot \hat{\mathbf{e}})^2$ gives an identity residual of 0.06672 at $y = 1$. This negative control illustrates why isotropy is a load-bearing hypothesis.

The current rerun passes six tidal tests, ten parent-action tests, six existing exact-inverse-kernel tests and two elliptic-action regression tests: 24 distinct tests, all with exit status zero. The parent checks include a finite-difference first variation. They do not collectively certify a relativistic field theory. The numerical environment is Python 3.9.6, NumPy 1.26.2, SciPy 1.11.4 and SymPy 1.14.0. The companion manifest records runtime, repository state and input/output checksums; the adjacent reproduction note gives commands. No astronomical observations were fitted or analyzed.

6 What a falsification would require

The first practical calculation is a fully nonlinear solution of (7) with a known external source, testing how the relation changes as the filtered internal field ceases to be small. The host system must also calibrate s_e : y is an inverse-kernel parameter, not automatically the measured physical external acceleration divided by a_0 . Milgrom’s external-boundary analysis distinguishes those fields in a general host [2]. A meaningful data test then needs joint identifiability of D and Q_2 , including ordinary tides, source shape and nuisance parameters. Increasing ξ can suppress both amplitudes together; the ratio alone does not exclude that escape.

7 Scope, priority and the unfinished relativistic problem

The proposition is an analytic consequence of the specified static action, its constant-background linearization and isotropic angular averages. It is not inferred from 324 numerical successes. Those checks do not test nonlinear source corrections, nonspherical systems or varying external fields. The exactly zero-background limit is singular in the constitutive tangent; finite central curvature at nonzero background does not settle strong coupling or loss of rank at zero field.

There is no physical spacetime metric in (4). This paper derives neither $\Phi = \Psi$ nor lensing, PPN parameters, a two-tensor Dirac count, covariant matter conservation, gravitational-wave speed, perturbative stability or FLRW evolution. Elliptic potentials are not proof that a covariant completion has no hidden mode or instantaneous physical channel. Both a_0 and ξ are inputs; no cosmological acceleration-scale relation is derived. The exact exponential constitutive choice is retained, but the stronger original AQUAL target is not fulfilled by this finite-filter action.

The explicit functional architecture has prior art [3]. The central filtered identity was not identified in the inspected source passages or the bounded repository/web searches documented with this draft. That is not evidence of global priority. The defensible contribution is an explicitly derived and reproducibly checked central-response corollary for this specialization. No claim of a first discovery is made.

8 Conclusion

The external-field anisotropy emphasized by Banik and Zhao survives isotropic smoothing in a constrained form: changing the filter amplitude cannot independently change the two central tidal coefficients in the stated linear regime. This is a concrete, conditional prediction from one action, not a fitted pair of unrelated signals. Its scientific value will be decided by nonlinear corrections and joint observational identifiability. The static identity is established under its assumptions; the full theory remains open.

References

- [1] I. Banik and H. Zhao, *The External Field Dominated Solution In QUMOND & AQUAL: Application To Tidal Streams*, [arXiv:1509.08457v2](#), 4 May 2018 revision; equations (22)–(25), (36).
- [2] M. Milgrom, *Quasi-linear formulation of MOND*, [arXiv:0911.5464v2](#), 2 March 2010 revision; equations (3)–(6) and section 5.
- [3] M. Milgrom, *Generalizations of quasilinear MOND (QUMOND)*, [arXiv:2305.01589v2](#), 4 October 2023 revision; section II.B and equations (6)–(7).
- [4] A. Hees, W. M. Folkner, R. A. Jacobson and R. S. Park, *Constraints on MOND theory from radio tracking data of the Cassini spacecraft*, [arXiv:1402.6950v2](#), 29 April 2014 revision; equations (5b) and (6).

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